Literature-Implied Answer: Barycentric Spaces Have Sharp Metric Cotype

Run `fa_banach_001`

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Status. `literature_implied_answer (partial subcase)`. Later literature settles the sharp scaling-parameter problem for q-barycentric metric spaces, but not for all finite-cotype Banach spaces.

Source question. Giladi–Mendel–Naor, *Improved bounds in the metric cotype inequality for Banach spaces*, arXiv:1003.0279, asks whether the metric cotype inequality for every Banach space of Rademacher cotype q can be proved with the sharp scaling parameter $m \asymp n^{1/q}$. Their Theorem 1 gives the general bound $m \lesssim n^{1+1/q}$, and the introduction identifies L_1 and the Schatten trace class S_1 as basic open cotype-2 cases.

Supporting theorem. Eskenazis–Mendel–Naor, *Nonpositive curvature is not coarsely universal*, arXiv:1808.02179, proves Theorem "q-barycentric implies sharp metric cotype q". If (X, d_X) is q-barycentric with constant β, then for every $n \in \mathbb{N}$, every even m, and every $f : \mathbb{Z}_{2m}^n \to X$,

$$\left(\sum_{i=1}^{n}\sum_{x} d_X(f(x+me_i), f(x))^q\right)^{1/q} \le (4n^{1/q} + \beta m)\left(2^{-n}\sum_{\varepsilon\in\{-1,1\}^n}\sum_{x} d_X(f(x+\varepsilon), f(x))^q\right)^{1/q}.$$

Consequently, $m \ge \beta^{-1}n^{1/q}$ gives metric cotype q with constant $\Gamma \lesssim \beta$, i.e. sharp scaling. The same paper proves that this sign-edge formulation is equivalent, up to universal constants, to the original ℓ_∞-edge formulation used in arXiv:1003.0279.

Scope and limitations. This is a positive partial answer for q-barycentric targets, including CAT(0)/Hadamard spaces and q-uniformly convex Banach spaces. It is not a solution of the main all-Banach problem: arXiv:1808.02179 explicitly states that whether every Banach space of Rademacher cotype q has sharp metric cotype q remains a fundamental open problem. In particular, this packet does not settle the highlighted L_1 or S_1 cases from arXiv:1003.0279.

Why this is an implication rather than an exact answer. The supporting paper is not framed as an answer to arXiv:1003.0279's all-Banach question. It cites the earlier metric-cotype framework, records the same remaining Banach-space obstruction, and proves a sharp theorem for a different structural class. Thus the correct durable status is a literature-implied partial subcase.

References.

- O. Giladi, M. Mendel, and A. Naor, *Improved bounds in the metric cotype inequality for Banach spaces*, arXiv:1003.0279.
- A. Eskenazis, M. Mendel, and A. Naor, *Nonpositive curvature is not coarsely universal*, arXiv:1808.02179.